

An Empirical Study of Vision-Language Model Fine-tuning for Radiology Report Generation: Why Retrieval Baselines Win on Small-Scale Medical Benchmarks

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Abstract

We conduct a systematic empirical study of vision-language model (VLM) fine-tuning for radiology report generation on IU-Xray, comparing seven approaches: LoRA fine-tuning, classifier-aligned prompting (CAT), visual feature fusion (VFF), MIMIC-CXR text-only pre-training, disease-hint prompting, nearest-neighbor retrieval, and an Oracle baseline. All fine-tuning methods plateau at BLEU-4 = 6.6–9.0; a zero-training nearest-neighbor retrieval baseline achieves BLEU-4 = 9.08, exceeding every learned model. Clinical metrics tell a different story: VFF achieves the best CheXbert-14 micro-F1 (54.17) and RadGraph scores among fine-tuned methods despite a slightly lower BLEU than baseline, suggesting that lexical overlap metrics systematically undervalue visually-grounded improvements on this benchmark. We trace these patterns to the dominance of the language prior on a 2,069-sample training set, gradient competition between language-side LoRA and visual-side adapters, and the structural homogeneity of IU-Xray reports. We argue that retrieval baselines and clinical metrics should be standard reporting practice on small-scale medical generation benchmarks.

1. Introduction

Automated radiology report generation on IU-Xray (Demner-Fushman et al., 2016) has reported steady BLEU-4 progress from 16.5 (R2Gen, Chen et al., 2020) to over 19 (SERPENT-VLM, Kapadnis et al., 2024). The emergence of strong general-purpose VLMs (Qwen2-VL, LLaVA, InternVL) suggests an attractive alternative to task-specific architectures: fine-tune a pretrained VLM with LoRA on the target medical dataset, optionally augmented with classifier hints or domain pre-training. This direction reuses large-scale general pre-training and avoids the cost of training task-specific models from scratch.

This work asks a different question. Rather than proposing a new method, we systematically investigate whether VLM fine-tuning on IU-Xray reliably improves over trivial baselines. We hold the base model (Qwen2-VL-7B) and dataset (IU-Xray, R2Gen split) fixed and compare seven approaches: pure LoRA, prompt-channel enhancement (CAT), visual-channel enhancement (VFF), text-only domain pre-training, plus nearest-neighbor retrieval and an Oracle reference.

Our findings are largely negative for fine-tuning. Every learned method matches or underperforms the no-hint baseline on BLEU-4; a 30-second nearest-neighbor retrieval baseline exceeds all of them. Diagnostic experiments suggest this is structural rather than a tuning artifact: zero-shot Qwen2-VL emits the same “cardiomegaly is noted...” template across visually distinct images, indicating the language prior dominates the visual signal. A subtler finding emerges from clinical metrics: VFF, which underperforms baseline on BLEU, achieves the best CheXbert-14 and RadGraph scores among fine-tuned methods—suggesting that BLEU on IU-Xray systematically rewards template matching over visual grounding.

Contributions. (1) A controlled comparison of seven radiology report generation methods on a fixed setup with both lexical and clinical metrics. (2) Mechanistic analysis of why prompt-channel and visual-channel enhancements fail at this data scale. (3) An empirical case for treating NN retrieval as a mandatory baseline and

for supplementing BLEU with clinical metrics.

2. Experimental Setup

Task and data. Given dual-view chest X-rays, generate the free-text findings paragraph. IU-Xray with R2Gen split: 2,069 train / 296 val / 590 test. A defining property: 92.6% of training reports contain normal-finding keywords, skewing the learning signal toward a small set of templates.

Base model. Qwen2-VL-7B-Instruct fine-tuned with LoRA (rank 16, alpha 32, target modules: q/k/v/o_proj, gate/up/down_proj). bfloat16 training on a single NVIDIA A100. Beam search with beam size 4 for inference.

Auxiliary classifier. DenseNet-121 initialized from torchxrayvision weights, fine-tuned on IU-Xray for 10-class multi-label classification (test macro-F1 = 0.221).

Metrics. Lexical: BLEU-1/2/3/4, ROUGE-L, METEOR. Clinical: CheXbert-5/14 F1 (micro and macro), RadGraph F1 (Simple and Partial). We additionally report the number of unique outputs across the 590 test samples as a diagnostic for mode collapse.

3. Methods

(1) LoRA Baseline. No-hint LoRA fine-tuning of Qwen2-VL-7B on IU-Xray. Reference point for all enhancement methods.

(2) CAT (Classifier-Aligned Training). DenseNet predictions thresholded into a comma-separated hint string, injected into the prompt during both training and inference to avoid train-test distribution mismatch.

(3) VFF (Visual Feature Fusion). Penultimate-layer DenseNet features (1024-d) fused with Qwen2-VL's vision merger output via gated cross-attention. The gate $g = \tanh(w)$ is initialized to zero, providing a lower-bound guarantee: the model starts identical to the no-hint baseline.

(4) MIMIC Text-Only Pre-training. LoRA pre-training of Qwen2-VL's language modules on 148K MIMIC-CXR findings paragraphs (three text-to-text tasks: findings→impression, paragraph continuation, hint→report) for 2 epochs, followed by IU-Xray fine-tuning. Image modality unused at the pre-training stage due to hardware constraints (full MIMIC images exceed 130GB).

(5) Disease-Hint. LoRA Baseline with GT-extracted training hints, DenseNet-predicted inference hints, 30% hint dropout, $2\times$ abnormal oversampling, and randomized prompt phrasings. Our primary configuration before the systematic comparison reported here.

(6) NN Retrieval. Zero-training baseline. For each test image, return the GT report of the training image with highest cosine similarity in DenseNet penultimate feature space. Runs in ~ 30 seconds on the full test set.

(7) Oracle. GT-leaked baseline: disease keywords extracted from the test sample's ground-truth report are injected into the prompt. Sanity check only; we argue against treating it as a method comparison target.

4. Results

Table 1. Main results on the IU-Xray test set (590 samples). Full table including BLEU-1/2/3, ROUGE-1/2, and CheXbert macro-F1 / RadGraph-Simple is reported in the full version.

Method	BLEU-4	ROUGE-L	METEOR	Unique	CheXb-5 mF1	CheXb-14 mF1	RG-Partial
<i>Fine-tuning methods</i>							
LoRA Baseline	8.48	30.41	23.95	30	0.00	51.63	33.15
Disease-Hint	~ 9.0	—	—	—	—	—	—

Method	BLEU-4	ROUGE-L	METEOR	Unique	CheXb-5 mF1	CheXb-14 mF1	RG-Partial
CAT	6.63	27.50	20.19	29	3.10	53.66	32.87
VFF	8.39	29.81	23.23	43	0.00	54.17	34.36
MIMIC Pre-training	6.80	27.15	20.02	17	10.84	36.35	30.79
<i>Retrieval / Reference</i>							
NN Retrieval (zero training)	9.08	27.62	—	—	25.33	41.45	30.29
Oracle (GT leak)	16.14	35.87	32.28	—	23.78	54.51	40.19

Observation 1: Fine-tuning methods plateau in a narrow band. BLEU-4 spans 6.63–9.0 across all five learned methods (CAT 6.63, MIMIC 6.80, VFF 8.39, LoRA 8.48, Disease-Hint ~9.0), despite substantial architectural differences. Elaborate enhancements either match or underperform the no-hint baseline. This pattern is consistent across BLEU-1/2/3, ROUGE, and METEOR (Table 1).

Observation 2: Zero-training retrieval exceeds every fine-tuned method on BLEU. NN Retrieval achieves BLEU-4 = 9.08, exceeding LoRA Baseline (8.48), VFF (8.39), CAT (6.63), and MIMIC (6.80). This replicates and strengthens Liu et al. (2019)'s observation in the modern VLM setting: the gap between learned generation and trivial retrieval has not closed despite five years of architectural progress.

Observation 3: Clinical metrics rank methods differently than BLEU. VFF achieves the best CheXbert-14 micro-F1 (54.17) and RadGraph scores among fine-tuned methods, exceeding LoRA Baseline (51.63, 33.15) despite slightly lower BLEU. NN Retrieval dominates CheXbert-5 (micro-F1 25.33 vs. 0.00 for LoRA and VFF). The LoRA Baseline's CheXbert-5 micro-F1 of 0.00 is a striking diagnostic: despite fluent BLEU-competitive text, the fine-tuned model produces essentially no correct clinical entities on the five most clinically critical CheXpert categories.

Output diversity. LoRA Baseline produces 30 unique outputs across 590 test samples (mode collapse); VFF produces 43 (the most diverse fine-tuned method); MIMIC pre-training collapses to 17 (worst). The unique-output count correlates with clinical performance, not with BLEU.

5. Failure Mechanism Analysis

Language prior dominance. Zero-shot Qwen2-VL with few-shot prompting outputs the same template—“cardiomegaly is noted, with the heart occupying more than 50% of the thoracic cavity”—on 4 out of 5 inspected samples, all of which have normal ground-truth reports. The model is not looking at the image; it is emitting a high-likelihood medical template. LoRA fine-tuning compresses this prior into IU-Xray's dominant “the heart is normal in size... the lungs are clear” template. The LoRA Baseline's CheXbert-5 micro-F1 of 0.00 is the direct numerical signature of this mechanism.

CAT: noisy hints and shortcut learning. DenseNet hint accuracy is approximately 55% per token. With 2,069 training samples, LoRA cannot learn the noisy-hint→correct-description mapping. A diagnostic experiment confirms shortcut learning: feeding the same image with different hints produces nearly identical outputs—the model uses hint presence as a context signal but ignores hint content.

VFF: gradient competition. The VFF gate converges to 0.0327 across multiple optimizer interventions ($10\times$ / $100\times$ gate learning rate, decoupled optimizer, adapter warmup). The cross-entropy objective can be driven down either by improving visual grounding (VFF pathway) or by compressing the LM into training-distribution templates (LoRA pathway); on 2,069 samples, the latter pathway dominates the optimization budget. The clinical-metric result is residual evidence that VFF is not failing—just out-competed by an easier pathway that BLEU rewards more.

MIMIC pre-training: template memorization. 148K MIMIC-CXR text reports cause the model to over-memorize “the heart is normal in size...”, which downstream visual fine-tuning on 2,069 samples cannot displace. Mode collapse to 17 unique outputs. The mechanism is the language-prior diagnosis amplified: more template prior before visual exposure makes the visual signal even less competitive.

Oracle is not a real upper bound. Oracle's BLEU-4 = 16.14 reflects template substitution of leaked GT keywords, not vision-grounded capability. Oracle's CheXbert-14 micro-F1 (54.51) is only marginally above VFF's (54.17), the expected pattern if Oracle is mostly a keyword-substitution shortcut rather than genuine improvement. A defensible non-cheating upper bound on this setup is likely in the BLEU-4 = 10–12 range, consistent with the gap to task-specific architectures.

6. Discussion

Scope of claims. Our results are scoped to small-scale medical generation (IU-Xray, 2,069 train) without large-scale medical pre-training. They do not establish that VLM fine-tuning is generally ineffective for medical report generation—LLaVA-Med with 213K MIMIC pre-training reaches BLEU-4 = 18.6—nor that retrieval is preferable for clinical deployment (retrieval returns another patient's exact report, unsuitable for production).

Implications for benchmark design. (1) NN retrieval should be reported as a standard baseline on IU-Xray-scale benchmarks; methods that fail to exceed it should not be reported as positive results. (2) Clinical metrics should accompany lexical metrics—our VFF result shows the two can rank methods differently, with clinical metrics rewarding genuine visual grounding that BLEU undervalues. (3) Oracle-style ground-truth leaks should not be treated as upper bounds.

Limitations. Single-seed evaluation (no variance estimation), single base model (Qwen2-VL-7B), single dataset (IU-Xray), no human evaluation, Disease-Hint clinical metrics incomplete.

Future directions. Retrieval-augmented generation that edits retrieved reports rather than generating from scratch; cleaner evaluation tasks (VQA, anatomy grounding) without template-matching confounds; pre-trained medical VLMs (LLaVA-Med, MAIRA-2) as the starting checkpoint rather than text-only pre-training substitutes.

7. Conclusion

On IU-Xray with Qwen2-VL-7B, every fine-tuning approach we tested plateaus at BLEU-4 = 6.6–9.0; a 30-second nearest-neighbor retrieval baseline achieves 9.08, exceeding all of them. Clinical metrics rank methods differently than BLEU, with VFF the strongest fine-tuned method on CheXbert-14 and RadGraph despite a slightly lower BLEU. The patterns trace to language prior dominance, gradient competition, and structural homogeneity of the benchmark's templated reports. The gap between learned generation and trivial retrieval—first noted by Liu et al. in 2019—has not closed despite five years of architectural progress. Retrieval baselines and clinical metrics should be standard reporting practice on small-scale medical generation benchmarks.

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